Assignment #1:

Total Points: 100

Office location: CAED 210H

Instructions

Collaboration Policy: For "Individual assignments," Collaboration is expected within the limits of discussing concepts and problems. However, each student must produce his/her own solution to the problems. For "Group assignments," each student should have specific contributions to the homework –by default, the instructor assumes equal contributions; if any disagreement between team members about the contributions of individuals, please talk to the instructor to initiate a peer evaluation process.

ChatGPT Policy: In general, please be transparent if you use ChatGPT and highlight the parts of the homework generated by ChatGPT. When you choose to use ChatGPT to provide some answers, please *1) use an online document to save the ChatGPT sessions that helped you produce the answer*; 2*) critically review the answers generated by ChatGPT, highlight the parts that you found that ChatGPT's answer needs improvements*.

General Expectations and Requirements for Homework:

- Please use considerable and uniform font sizes throughout the document to maintain consistency of the document. You may choose to highlight subheadings with either a bold or underlined feature.
- Please use bullet points wherever possible to make the answers clear and easy to follow by an educated reader.
- Please do not forget to reference additional data, hyperlinks or literature used as evidence or background information to support your claims and solutions in the document. Please list those references below your answer or at the end of the document.
- Please refer to the textbook and provide descriptive answers wherever possible.
- <u>Please communicate with the instructor to clarify questions about the homework description BEFORE the submission</u>; after the homework submission deadline, the students are responsible for the point losses due to different ways of interpreting the homework requirements.

All homework submissions should be submitted electronically on Canvas.

Student Name:_____
ID: _____

This assignment aims to deepen your understanding of the key challenges in the construction industry and enhance your skills in data visualization using Python in Google Colab. You will analyze real-world data, identify challenges, and propose potential solutions.

1. Essay Question (50 points)

What are the pressing challenges faced by the architecture/construction industry today? Please include necessary data (with citations) to indicate why these are challenges, and discuss how we could possibly address those challenges. Please adhere to academic standards. Directly copying from other sources without proper citations will be considered plagiarism!

2. Coding Practice (50 points)

Use Python in Google Colab to visualize the challenges in the architecture/construction industry. Include **at least three data-driven visualizations** to support your discussion.

Hint: You can use a hypothetical or publicly available dataset related to construction costs, delays, labor statistics, etc.

For the coding practice, please insert the generated figures into this submission and upload your Google Colab notebook containing all the visualizations.

Here is one visualization example:

```
# Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Example dataset loading
data = pd.read_csv('path_to_your_dataset.csv')

# Create a bar chart showing the cost overruns in various construction
projects across different states or countries
plt.figure(figsize=(10, 6))
plt.bar(data['Project'], data['Cost_Overrun'], color='red')
plt.xlabel('Project')
plt.ylabel('Cost Overrun ($)')
plt.title('Cost Overruns in Construction Projects')
plt.show()
```

<u>Final Question:</u> (0 points)
How much time did it take you to finish this assignment? Please specify the number of hours.

Good Luck!!